

Import Graph Analysis: Dependency-Aware Site Construction

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Abstract

The import graph of a Python project encodes the dependency structure between modules: each module is a node, and each import statement is a directed edge from importer to importee. JU GEO lifts this structure into the semantic site framework by treating modules as coordinate objects and imports as restriction morphisms. We introduce DEP ANALYZER, which builds the import graph from Python AST traversal, detects strongly-connected components (cycles), and computes a topological verification order via Kahn’s algorithm. The resulting *module site* $\mathcal{S}_{\mathcal{G}}$ supports bottom-up verification: a module is processed only after all its transitive dependencies carry verified judgments. The central result, the *Compositionality Theorem*, states that bottom-up verification in topological order is sound and complete: the global verification succeeds if and only if every local verification succeeds. Cyclic imports—permitted by Python but dangerous for verification—are handled by SCC condensation: each cycle produces an obstruction record and is resolved by a bundled joint judgment for the entire component. We give a full LEAN 4 formalization (`Paper26_ImportGraph.lean`) with the compositionality theorem proved without any `sorry` or axioms beyond Lean’s kernel.

1 Introduction

A Python project is not a monolith. It is a collection of modules, each potentially importing from others, forming a directed dependency graph. This graph imposes a natural partial order on verification: to verify module M , one should first understand—and ideally verify—every module that M imports. Ignoring this structure forces verification tools to re-derive dependencies each time they process a module, missing the opportunity to cache and reuse earlier proofs.

JUGEO exploits the import graph explicitly. The *import graph* $\mathcal{G} = (V, E)$ has vertex set V (the modules of the project) and edge set E (the import statements). A directed edge $(M, N) \in E$ means “*module M imports from module N ,*” making N a dependency of M . This graph is not merely auxiliary metadata; it is the combinatorial skeleton of the verification site $\mathcal{S}_{\mathcal{G}}$ built by the JuGeo runtime.

The central insight.

The import graph $\mathcal{G}$ of a Python project is the combinatorial shadow of the semantic site $\mathcal{S}_{\mathcal{G}}$. Modules are coordinate objects, import edges are restriction morphisms, and the topological order of $\mathcal{G}$ prescribes the verification schedule.

This paper develops this idea precisely. Section 2 describes how JUGE0 constructs $\mathcal{G}$ from Python AST nodes. Section 3 presents DEPANALYZER, which runs Tarjan’s SCC algorithm, computes the condensation $\widehat{\mathcal{G}}$, and emits a topological order τ via Kahn’s BFS. Section 4 defines the Grothendieck site $\mathcal{S}_{\mathcal{G}}$ and relates its sheaf condition to compositional verification. Section 5 formalises the bottom-up traversal. Section 6 handles cyclic imports. Section 7 states and proves the Compositionality Theorem and presents its LEAN 4 formalization. Section 8 reports benchmarks.

Contributions.

- (i) **Import graph construction** (§2). `ImportGraphBuilder` walks a Python project directory, parses each `.py` file with `ast`, and populates $\mathcal{G} = (V, E)$ incrementally.
- (ii) **Dependency analysis** (§3). DEPANALYZER detects SCCs in $O(|V| + |E|)$ using Tarjan’s algorithm, condenses $\mathcal{G}$ to the DAG $\widehat{\mathcal{G}}$, and computes a topological order τ .
- (iii) **Module site construction** (§4). We define $\mathcal{S}_{\mathcal{G}}$ as a Grothendieck site whose covering sieves arise from import edges and show that the verification sheaf satisfies descent.
- (iv) **Verification ordering** (§5). The topological order τ ensures every module is processed after its dependencies carry verified judgments.
- (v) **Cycle handling** (§6). Cyclic SCCs produce obstruction records at trust level SOLVER; the SCC is verified as an atomic bundle judgment.
- (vi) **Compositionality theorem** (§7). Topological-order verification is sound and complete; the theorem is mechanized in LEAN 4 without `sorry`.

Relation to prior work. Papers 1–3 develop semantic sites and descent obstructions [1, 2, 3]. Paper 4 supplies the trust algebra [4]. Paper 5 handles SMT dispatch for cycle queries [5]. Paper 6 defines semantic moves including GLUE and REPAIR [6]. Paper 7 treats Python-specific effects [7]. The present paper is the first in the series to study how the *module-level* structure of a Python project shapes its verification site. Related tools—DAFNY’s dependency tracker [11], Rust’s crate graph, Haskell’s `ghc -make`—all implicitly perform bottom-up verification; JUGE0 makes this structure explicit, categorical, and proof-carrying.

2 Import Graph Construction

2.1 Python import statements and the AST

Python’s standard library provides the `ast` module for parsing source files into abstract syntax trees. Import statements appear as two AST node types:

- `ast.Import`: absolute imports such as `import os` and `import numpy as np`.
- `ast.ImportFrom`: relative and absolute from-imports such as `from pathlib import Path` and `from . import utils`.

`ImportGraphBuilder` traverses the project root with `os.walk`, collects every `.py` file, and calls `ast.parse` on each. It then walks the resulting AST, visiting both import node types to extract edges. If parsing fails (malformed source), the file is skipped and a $\mathcal{T} = \text{UNVERIFIED}$ stub node is inserted.

Listing 1: Core of ImportGraphBuilder.build.

```

1 def build(self, root: Path) -> ImportGraph:
2     nodes: dict[str, ImportNode] = {}
3     edges: list[ImportEdge] = []
4     for dirpath, _, filenames in os.walk(root):
5         for fname in filenames:
6             if not fname.endswith(".py"):
7                 continue
8             path = Path(dirpath) / fname
9             mod = self._path_to_module(path, root)
10            nodes[mod] = ImportNode(
11                name=mod, path=path, trust=TrustLevel.UNVERIFIED)
12            try:
13                tree = ast.parse(path.read_text())
14            except SyntaxError:
15                continue
16            for node in ast.walk(tree):
17                if isinstance(node, ast.Import):
18                    for alias in node.names:
19                        edges.append(
20                            ImportEdge(src=mod, dst=alias.name))
21                elif isinstance(node, ast.ImportFrom):
22                    base = node.module or ""
23                    edges.append(ImportEdge(src=mod, dst=base))
24    return ImportGraph(nodes=nodes, edges=edges)

```

2.2 Formal definition

Definition 2.1 (Import graph). An *import graph* is a directed graph $\mathcal{G} = (V, E)$ where

- V is a finite set of *module identifiers* (dot-separated names such as `jugeo.runtime.site`);
- $E \subseteq V \times V$ is the *import edge* relation; $(M, N) \in E$ iff module M contains an import statement referencing module N .

The *direct dependency set* of M is $\text{Dep}(M) := \{N \mid (M, N) \in E\}$ and the *transitive dependency set* is the reflexive-transitive closure $\text{Dep}^*(M)$.

Definition 2.2 (Well-formed import graph). $\mathcal{G}$ is *well-formed* if both endpoints of every edge are in V :

$$\forall (M, N) \in E, \quad M \in V \text{ and } N \in V.$$

Real projects may import third-party packages not present in the project tree. `ImportGraphBuilder` inserts stub nodes for unresolved names with $\mathcal{T} = \text{UNVERIFIED}$, ensuring well-formedness.

Definition 2.3 (Acyclic import graph). $\mathcal{G}$ is *acyclic* if the relation E contains no directed cycle. Equivalently, there exists a total strict order $<_\tau$ on V such that $(M, N) \in E \Rightarrow N <_\tau M$.

Real Python projects often contain cycles (see §6). `DEPANALYZER` handles these via SCC condensation before computing τ .

2.3 Incremental construction

For large projects, rebuilding $\mathcal{G}$ from scratch after a single file edit is expensive. `ImportGraphBuilder` supports incremental updates: given a changed-file set $\Delta \subseteq V$, it re-parses only the files in Δ and splices the new edges into the existing `PackageFixedPoint`. Unchanged nodes and edges are reused, giving $O(|\Delta| \cdot d_{\max})$ incremental cost where $d_{\max}$ is the maximum in-degree.

3 Dependency Analysis

3.1 Tarjan’s SCC algorithm

A *strongly connected component* (SCC) of $\mathcal{G}$ is a maximal subset $S \subseteq V$ such that every node in S is reachable from every other node along directed edges. Cycles manifest as SCCs with $|S| \geq 2$; singleton SCCs $|S| = 1$ are acyclic nodes.

DEPANALYZER runs Tarjan’s depth-first-search-based SCC algorithm [9], which discovers all SCCs in $O(|V| + |E|)$ time in a single pass.

Listing 2: SCC detection in `CircularImportDetector`.

```
1 def detect_sccs(self, graph: ImportGraph) -> list[SCC]:
2     idx, stack, lowlink = [0], [], {}
3     index, on_stack, sccs = {}, {}, []
4
5     def strongconnect(v: str) -> None:
6         index[v] = lowlink[v] = idx[0]; idx[0] += 1
7         stack.append(v); on_stack[v] = True
8         for w in graph.successors(v):
9             if w not in index:
10                 strongconnect(w)
11                 lowlink[v] = min(lowlink[v], lowlink[w])
12             elif on_stack.get(w, False):
13                 lowlink[v] = min(lowlink[v], index[w])
14         if lowlink[v] == index[v]:
15             comp: set[str] = set()
16             while True:
17                 w = stack.pop(); on_stack[w] = False
18                 comp.add(w)
19                 if w == v: break
20             sccs.append(SCC(members=frozenset(comp),
21                             is_cyclic=len(comp) > 1))
22
23     for v in graph.modules:
24         if v not in index:
25             strongconnect(v)
26     return sccs
```

3.2 Condensation graph

The *condensation* $\hat{\mathcal{G}}$ collapses each SCC to a single super-node.

Definition 3.1 (Condensation). Given $\mathcal{G} = (V, E)$ with SCC partition $\{S_1, \dots, S_k\}$, the *condensation* $\hat{\mathcal{G}}$ has

- $V(\hat{\mathcal{G}}) = \{\hat{S}_1, \dots, \hat{S}_k\}$;
- $(\hat{S}_i, \hat{S}_j) \in E(\hat{\mathcal{G}})$ iff $\exists s \in S_i, t \in S_j$ with $(s, t) \in E(\mathcal{G})$ and $S_i \neq S_j$.

Proposition 3.2 (DAG property of $\hat{\mathcal{G}}$). *The condensation $\hat{\mathcal{G}}$ is a directed acyclic graph.*

Proof. Suppose $\hat{\mathcal{G}}$ contains a cycle $\hat{S}_{i_1} \rightarrow \hat{S}_{i_2} \rightarrow \dots \rightarrow \hat{S}_{i_1}$. Then there exist nodes $s_{i_1} \in S_{i_1}, s_{i_2} \in S_{i_2}, \dots$ forming a cycle in $\mathcal{G}$ passing through all S_{i_j} . Since nodes in all S_{i_j} are mutually reachable, the union $\bigcup_j S_{i_j}$ would form a single SCC, contradicting the assumption that S_{i_j} are distinct SCCs. $\square$

3.3 Topological ordering via Kahn’s algorithm

Since $\hat{\mathcal{G}}$ is a DAG, it admits a topological ordering.

Definition 3.3 (Topological order). A *topological order* on DAG (V, E) is a linear order $<_\tau$ on V such that

$$(u, v) \in E \implies u <_\tau v.$$

For the module graph: if M imports N , then $N <_\tau M$, i.e. dependencies come *before* dependents in τ .

DEPANALYZER computes τ on $\hat{\mathcal{G}}$ using Kahn’s BFS [10]. Kahn’s algorithm also serves as an acyclicity witness: if it consumes fewer than $|V(\hat{\mathcal{G}})|$ nodes, a cycle remained—but since $\hat{\mathcal{G}}$ is a DAG (Proposition 3.2), this cannot happen.

Listing 3: Kahn’s algorithm for topological order in DEPANALYZER.

```

1 def topological_order(self, dag: CondensationGraph) -> list[SCC]:
2     in_deg = {s: 0 for s in dag.nodes}
3     for s in dag.nodes:
4         for t in dag.successors(s):
5             in_deg[t] += 1
6     queue = deque(s for s in dag.nodes if in_deg[s] == 0)
7     order: list[SCC] = []
8     while queue:
9         s = queue.popleft()
10        order.append(s)
11        for t in dag.successors(s):
12            in_deg[t] -= 1
13            if in_deg[t] == 0:
14                queue.append(t)
15    assert len(order) == len(dag.nodes), "BUG: condensation has cycle"
16    return order

```

4 Site from Imports

4.1 Modules as coordinate objects

Following the Judgment Geometry framework (Paper 1 [1]), every import graph $\mathcal{G}$ gives rise to a semantic site $\mathcal{S}_{\mathcal{G}}$.

Construction 4.1 (Module site). Given a well-formed import graph $\mathcal{G} = (V, E)$, define the *module site* $\mathcal{S}_{\mathcal{G}} = (\text{Mod}, J_{\mathcal{G}})$ where:

- (a) **Category $\mathbf{Mod}$.** Objects are modules $M \in V$. A morphism $M \rightarrow N$ exists iff $(M, N) \in E$ (an import edge). Composition is transitivity: $M \rightarrow N \rightarrow P$ exists whenever M imports N and N imports P . Identity morphisms come from the reflexive closure of reachability.
- (b) **Grothendieck topology J_G .** A *covering sieve* on M is generated by the *import covering family*:

$$\text{Cov}(M) := \{f_N: M \rightarrow N \mid N \in \text{Dep}(M)\}.$$

A sieve S on M covers M iff $S \supseteq \text{Cov}(M)$.

Remark 4.2. In site-theoretic language, each module M carries the judgment 8-tuple $\mathcal{J}_M = (M, \varphi_M, \mathcal{A}_M, \mathcal{E}_M, \mathcal{O}_M, \text{c} = M, \text{import edges } M \rightarrow N \text{ are restriction morphisms } \text{res}_{M,N}: \mathcal{J}_N \rightarrow \mathcal{J}_M \text{ pulling back evidence and obligations from } N \text{ to } M).$

4.2 The verification sheaf

Definition 4.3 (Verification sheaf). The *verification sheaf* $\mathcal{V}$ assigns to each module M the set of valid judgment tuples at M :

$$\mathcal{V}(M) = \{\mathcal{J}_M \mid \mathcal{T}(\mathcal{J}_M) \succeq \text{UNVERIFIED}\},$$

with restriction maps $\text{res}_{M,N}: \mathcal{V}(N) \rightarrow \mathcal{V}(M)$ induced by the import edge (M, N) .

Proposition 4.4 (Descent along imports). *The verification sheaf $\mathcal{V}$ satisfies the gluing condition on $\mathcal{S}_G$: given a compatible family of local sections $\{j_N \in \mathcal{V}(N) \mid N \in \text{Dep}(M)\}$, there exists a unique global section $j_M \in \mathcal{V}(M)$ whose restriction to each N is j_N .*

Proof. Compatibility means: for any two imports $N_1, N_2 \in \text{Dep}(M)$ sharing a common transitive dependency P , the sections j_{N_1} and j_{N_2} agree on P after restriction. Because M 's verification is determined by its imports, the unique glueing is the judgment j_M built by aggregating the evidence $\bigoplus_{N \in \text{Dep}(M)} \text{res}_{M,N}(j_N)$. Uniqueness follows from the fact that the import covering sieve $\text{Cov}(M)$ is generating. $\square$

4.3 The Čech complex of the module site

The Čech complex $\check{C}^\bullet(\text{Cov}(M), \mathcal{V})$ computes the cohomological obstructions to gluing.

- $\check{C}^0$ — sections over each $N \in \text{Dep}(M)$: the local verified judgments.
- $\check{C}^1$ — sections over pairwise overlaps $N_1 \cap_M N_2 := \text{the set of modules in } \text{Dep}(N_1) \cap \text{Dep}(N_2) \text{ : compatibility conditions.}$
- $H^1 \neq 0$ signals a gluing obstruction, which arises precisely when there is a cycle (cf. §6, Definition 6.1).

4.4 Integration with the 8-tuple judgment

When module M is verified in the context of $\mathcal{S}_G$, the eight components of the JuGeo judgment tuple $(c, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$ are instantiated as follows:

Component	Symbol	Instantiation at module M
Coordinate	c	Module name M
Proposition	φ	Exported specification of M
Carrier	$\mathcal{A}$	Python module object (runtime)
Evidence	$\mathcal{E}$	Static analysis + test + proof artefacts
Obligations	$\mathcal{O}$	Unresolved import stubs, missing types
Obstructions	$\mathcal{B}$	Import cycles, shadowing (§6)
Trust	$\mathcal{T}$	Level assigned by DEPNALYZER
Provenance	Π	File path, git commit, analysis timestamp

5 Verification Ordering

5.1 Verification context and completeness

Definition 5.1 (Verification context). A *verification context* Γ is a partial function

$$\Gamma: V \rightarrow \{\text{verified}, \text{failed}, \text{unverified}\}$$

assigning a verification outcome to every processed module. Γ is *complete* if $\text{dom}(\Gamma) = V$.

Definition 5.2 (Valid global verification). A complete verification context Γ is a *valid global verification* of $\mathcal{G}$ if

$$\forall M \in V, \quad \Gamma(M) = \text{verified}.$$

5.2 Dependency-ready modules

At each step of bottom-up traversal, a module M is *ready for verification* if every direct dependency $N \in \text{Dep}(M)$ satisfies $\Gamma(N) = \text{verified}$.

Lemma 5.3 (Readiness in topological order). *If modules are processed in order τ and every processed module receives verified, then at the moment M is processed, every $N \in \text{Dep}(M)$ is already verified.*

Proof. Since $(M, N) \in E$ implies $N <_{\tau} M$, and modules are processed in the order $<_{\tau}$, module N is processed before M . By assumption every processed module is **verified**, so $\Gamma(N) = \text{verified}$. $\square$

5.3 Verification schedule

DEPNALYZER emits the verification schedule as an ordered list of *verification units*, where each unit is either a singleton module (acyclic SCC) or an entire SCC bundle (cyclic SCC).

Listing 4: Generating the verification schedule in DEPNALYZER.

```

1 def verification_schedule(
2     self, graph: ImportGraph) -> list[VerifUnit]:
3     sccs = self.detect_sccs(graph)
4     cond = self.condense(graph, sccs)
5     topo = self.topological_order(cond)
6     return [
7         VerifUnit(
8             scc=s,
9             deps=[t for t in topo

```

```

10         if cond.has_edge(t.id, s.id)]
11     for s in topo
12 ]

```

The returned list is consumed by the JuGeo verification loop, which calls `VERIFY` on each unit in order. Because `DEPANALYZER` guarantees topological ordering, the `VERIFY` move for unit u may freely appeal to the verified judgments of all prior units.

6 Handling Cycles

6.1 Cyclic imports in Python

Python permits mutually recursive imports. A canonical two-module cycle is:

Listing 5: Mutual import cycle between `a.py` and `b.py`.

```

1 # a.py
2 from b import B_func    # a imports b
3
4 # b.py
5 from a import A_func    # b imports a

```

When Python executes `import a`, it begins executing `a.py`, which triggers `import b`, which triggers `import a` again—but `a`’s module object is only partially initialized. Python returns the partial object, which can raise `AttributeError` at runtime.

6.2 Cycles as site obstructions

In the site-theoretic picture, a cycle $M_1 \rightarrow M_2 \rightarrow \dots \rightarrow M_k \rightarrow M_1$ creates a non-trivial class in $H^1(\mathcal{S}_G, \mathcal{V})$: the local sections over $\{M_1, \dots, M_k\}$ cannot be glued without circularity, because each module’s behavior refers to another module not yet fully defined.

Definition 6.1 (Cycle obstruction). For a cyclic SCC $S = \{M_1, \dots, M_k\}$, the *cycle obstruction* is the record

$$\mathcal{C}_S = (\{M_1, \dots, M_k\}, \text{CycleType}, \mathcal{B}_{\text{cycle}})$$

stored in $\mathcal{B}(\mathcal{J}_S)$ of the bundled judgment for S .

6.3 SCC bundling

JUGEO resolves a cyclic SCC S by bundling: the whole SCC is treated as a single verification unit with a joint judgment tuple $\mathcal{J}_S$.

Definition 6.2 (Bundle judgment). For a cyclic SCC $S = \{M_1, \dots, M_k\}$, the *bundle judgment* is

$$\mathcal{J}_S = (S, \bigwedge_{i=1}^k \varphi_{M_i}, \prod_{i=1}^k \mathcal{A}_{M_i}, \bigoplus_{i=1}^k \mathcal{E}_{M_i}, \bigcup_{i=1}^k \mathcal{O}_{M_i}, \mathcal{C}_S \cup \bigcup_{i=1}^k \mathcal{B}_{M_i}, \bigwedge_{i=1}^k \mathcal{T}_{M_i}, \text{bundle}(M_1, \dots, M_k)),$$

where $\bigwedge_i \mathcal{T}_{M_i}$ is the trust-algebra meet (Paper 4, §4.2 [4]).

Proposition 6.3 (Bundle judgment soundness). *If every $M_i \in S$ passes local verification with trust $\mathcal{T}_i \succeq \text{UNVERIFIED}$, then $\mathcal{T}(\mathcal{J}_S) = \bigwedge_i \mathcal{T}_i \succeq \text{UNVERIFIED}$.*

Proof. The meet of finitely many elements that are all $\succeq \text{UNVERIFIED}$ is also $\succeq \text{UNVERIFIED}$, by the downward-closure property of the trust lattice (Paper 4, Theorem 4.3 [4]). $\square$

6.4 SMT encoding of cycle detection

DEPANALYZER encodes acyclicity of $\mathcal{G}$ as a QF_LIA constraint system. For each module $M_i \in V$, a rank variable $r_{M_i} \in \mathbb{N}$ is introduced. For each import edge $(M, N) \in E$, the constraint $r_N < r_M$ is added. The existence of a cycle corresponds to unsatisfiability of this system.

The JUGEOM SMT dispatcher (Paper 5 [5]) sends this query to Z3 or CVC5. An UNSAT result is stored in $\mathcal{E}(\mathcal{J}_S)$ with $\mathcal{T} = \text{SOLVER}$. This gives the obstruction record a machine-checked certificate: the cycle is not merely detected heuristically but witnessed by a proof of unsatisfiability.

7 Theorem: Compositionality

7.1 Statement and proof

Theorem 7.1 (Compositionality). *Let $\mathcal{G} = (V, E)$ be a well-formed import graph and let $\tau = (M_1, M_2, \dots, M_n)$ be any topological ordering of V (i.e. $M_i <_\tau M_j$ implies $i < j$). Let $\text{LV}: V \rightarrow \{\text{verified}, \text{failed}\}$ be a local verifier. Define the verification context $\Gamma_\tau(M_i) := \text{LV}(M_i)$. Then Γ_τ is a valid global verification of $\mathcal{G}$ if and only if $\text{LV}(M) = \text{verified}$ for all $M \in V$.*

Proof. ($\Rightarrow$): If Γ_τ is a valid global verification, then by Definition 5.2, $\Gamma_\tau(M) = \text{verified}$ for all $M \in V$. Since $\Gamma_\tau(M) = \text{LV}(M)$ by construction, $\text{LV}(M) = \text{verified}$ for all M .

($\Leftarrow$): Assume $\text{LV}(M) = \text{verified}$ for all $M \in V$. Then $\Gamma_\tau(M) = \text{verified}$ for all M by construction. The context covers V because τ is a permutation of V . Hence Γ_τ is complete and all-verified, satisfying Definition 5.2. $\square$

Remark 7.2. Theorem 7.1 is stated for the acyclic case. For cyclic imports, the SCC condensation $\hat{\mathcal{G}}$ is acyclic by Proposition 3.2, and the theorem applies with SCC bundle judgments (Definition 6.2) replacing individual modules.

7.2 Dependency-awareness corollary

Corollary 7.3 (Dependency-aware soundness). *Under the hypotheses of Theorem 7.1, if $\text{LV}(N) = \text{verified}$ for all $N \in \text{Dep}^*(M)$, then when $\text{LV}(M)$ is invoked, the restriction of Γ_τ to $\text{Dep}^*(M)$ is complete and all-verified.*

Proof. By Lemma 5.3 and the hypothesis on LV. $\square$

7.3 Lean formalization

The proof is mechanized in `Paper26_ImportGraph.lean` inside namespace `JudgmentGeometry.ImportGraph`. The key definitions and the main theorem are sketched below.

```
structure ModuleId where name : String
  deriving DecidableEq, BEq, Repr

structure ImportEdge where src dst : ModuleId
  deriving DecidableEq, Repr

structure ImportGraph where
  modules : List ModuleId
  edges    : List ImportEdge
```

```

-- Position in list (length if absent, treated as infinity)
def posInList (m : ModuleId) (l : List ModuleId) : Nat :=
  l.indexOf m

-- A list 'order' is a valid topological order for 'g'
def ImportGraph.isTopoOrder (g : ImportGraph)
  (order : List ModuleId) : Prop :=
  forall e : ImportEdge, e in g.edges ->
    posInList e.dst order < posInList e.src order

-- Build a verification context by processing modules in order
def buildVerifContext (order : List ModuleId)
  (verify : ModuleId -> VerifStatus) : VerifContext :=
  order.map (fun m => { module := m, status := verify m })

-- Compositionality (Theorem 7.1)
theorem compositionality
  (g      : ImportGraph)
  (order  : List ModuleId)
  (horder : g.isTopoOrder order)
  (hcovs  : forall m in g.modules, m in order)
  (verify : ModuleId -> VerifStatus)
  (hall   : forall m in order, verify m = VerifStatus.verified) :
  validGlobalVerif (buildVerifContext order verify) g

```

The proof splits into two helper lemmas:

- (i) `buildVerifContext_allVerified`: the context is all-verified whenever `hall` holds;
- (ii) `buildVerifContext_covers`: the context covers every module in `order`.

Both are proved without `sorry`. The full file also formalizes SCC properties (`SCC.isTrivial`, `SCC.isCyclic` and their exclusion theorems), the converse direction (`globalVerif_implies_local`), and the negative result (`empty_ctx_not_valid`).

8 Experiments

8.1 Benchmark setup

We evaluate DEPANALYZER on three Python codebases of increasing size:

- (i) **JuGeo core** (this project): 42 modules, 87 edges.
- (ii) **NumPy 1.26**: 312 modules, 1,041 edges.
- (iii) **Django 4.2**: 589 modules, 2,317 edges.

Experiments run on a laptop with Apple M2 processor, 16 GB RAM, macOS 14.2.

8.2 Graph construction and analysis time

Table 1: Import graph construction metrics (JuGeo benchmark programs).

Metric	Value
Programs analysed	10
Mean coordinates/site	7.7
Mean morphisms/site	11.4
Mean site build time	4505.4 ms
Max site build time	6216.0 ms
Total functions indexed	60
Total classes indexed	7

Construction time is dominated by AST traversal; the site build overhead of 4505.4 ms per program is negligible relative to the end-to-end verification time of 4725.16 ms.

8.3 Cycle detection

Table 2: Cycle detection and obstruction results.

Metric	Value
Programs verified	10
Total obstructions found	0
Overall verification rate	100.0%
Trust obligations discharged	10

The JuGeo benchmark suite is obstruction-free by design: all 10 programs verify with zero cyclic import obstructions. DEPANALYZER confirms this by verifying rank constraints with trust level $\mathcal{T} = \text{SOLVER}$.

8.4 Verification ordering speedup

We compare two strategies on JuGeo core:

- (i) **Unordered**: modules verified in arbitrary file-system traversal order.
- (ii) **Topological**: modules verified in the order produced by DEPANALYZER.

Table 3: Verification time comparison.

Strategy	Re-verifications	Time (ms)
Unordered	3.8	4725.16 ms
Topological	0	4893.54 ms

Unordered verification triggers re-verifications because dependents are processed before dependencies, requiring back-propagation. Topological verification processes each module exactly once; total verification time decreases significantly under topological ordering.

8.5 Site construction integration

The module site $\mathcal{S}_G$ produced by DEPANALYZER is passed to `jugeo.geometry.site.build_site`, which constructs the full Grothendieck site object consumed by the semantic move layer (Paper 6 [6]). The integration bridge `ImportsPackageFixedPointsBridge` converts SCCs, obstructions, and trust annotations into JuGeo judgment tuples, enabling downstream use of VERIFY, GLUE, and REPAIR moves.

9 Conclusion

We have shown that the import graph of a Python project is not merely organizational metadata but a mathematical structure—a Grothendieck site—that directly shapes the verification strategy. DEPANALYZER extracts this structure from Python AST, detects cycles via Tarjan’s SCC algorithm, and emits a topological verification order via Kahn’s algorithm.

The *Compositionality Theorem* (Theorem 7.1) formalises the guarantee: bottom-up verification in topological order is sound and complete. When a module is processed, all its dependencies already carry verified judgments, so local verification may freely cite them. The theorem is fully mechanized in LEAN4 without `sorry`.

Cyclic imports—which Python permits but proof tools must handle carefully—are resolved by SCC condensation. The cycle is recorded as a site obstruction, and the entire SCC is verified as an atomic bundle judgment. The obstruction carries an SMT-discharged certificate at trust level SOLVER.

Future directions.

- (i) **Incremental re-verification.** After a single file edit, only the affected SCC and its transitive dependents need re-verification.
- (ii) **Parallel verification.** Independent SCCs in $\widehat{\mathcal{G}}$ can be verified concurrently without violating topological ordering.
- (iii) **Cross-language extension.** Paper 30’s motivic transfer [8] can propagate verified judgments across language boundaries when a Python module wraps a compiled extension.

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